Tomek Loboda

Full-Stack Data Scientist

Experienced data scientist, software engineer, and research scientist with competence in modeling and simulation, Bayesian inference, optimization, eye tracking, remote sensing, psycholinguistics, power systems, finance, ed-tech, epidemiology, and agriculture.

Independent, systematic, good communicator, pursues high quality and always seeks to simplify.

Contact

Address
Pittsburgh, PA

Phone
(412) 417 5610

Email
email@tomekloboda.net

Web
tomekloboda.net

GitLab
gitlab.com/ubiq-x

LinkedIn
linkedin.com/in/tomekloboda

Competence Areas

Computer science
Statistics
Modeling and simulation
Signal processing
Optimization
Eye-tracking
Remote sensing
Computational psycholinguistics
Educational technology
Psycholinguistics
Finance
Edge computing
Robotics and electronics
Epidemiology and agriculture

Skills

Software engineering
Statistical inference
Machine learning
Data engineering
Visualization
UI design
Full-stack Web development

Education

2014 — **PhD • Information Science**
University of Pittsburgh

2003 — **MS • Computer Science**
Wroclaw University of Science and Technology

Experience

From 2024-11 — **Senior Research Engineer**
Pearl Street Technologies • Pittsburgh

- Developing and evaluating power systems economics methodology (optimization and circuit theory; IEEE, ACTIVSg, ISO market data)
- Supporting transmission system impact studies for new generation resources

2023-09 – 2024-11 — **Machine Learning Engineer • Data Engineer**
Pittsburgh

- Financial modeling (e.g., time series forecasting, market regime detection)
- Developing trading strategies and building a trading system

2019-12 – 2023-06 — **Senior Research Software Engineer • Research Scientist**
Momacs Institute • University of Pittsburgh

- Building a software architecture comprising Earth observation, biophysical models, and other models, for sub-Saharan Africa rice systems
- Building end-to-end machine learning pipelines and performing statistical inference based on remote sensing and crop dynamics data
- Building a novel probabilistic relational agent-based modeling-and-simulation framework
- Simulating and investigating complex interacting systems
- Modeling multidisciplinary problems
- Building a database to support fast and automatic locale-based epidemiological model evaluation and comparison with focus on Covid-19
- Developing a multivariate time series clustering method (Wavelet transforms)
- Building a data collection and knowledge engineering system
- Leading software development and research efforts
- Liaising with clients; Mentoring

- **Datasets:** MSI and SAR imagery, geospatial, agriculture, climate, epidemiology, public health, population, synthetic population, administrative

Toolkit

Python • R • SAS • SQL
JavaScript • HTML • CSS
Java • C • C++ • bash

Postgres • PostGIS • TimescaleDB
FastAPI • Node.js
MLflow • Ray • Spark
FreeBSD • Linux • AWS
Dash • D3.js
Docker • LaTeX

Google Earth Engine • QGIS
SuiteSparse
EyeLink • Tobii • DSSAT

Python

numpy • scipy • pywt
pandas • polars • xarray
scikit-learn • statsmodels
pymc • numpyro • arviz
xgboost • torch • jax • keras
rasterio • shapely • geopandas
matplotlib • seaborn
folium • geemap
pyomo • cvxpy

Languages

English • Polish

Interests

Cycling
Piano
Photography and videography
Programming
Robotics
Science / Philosophy of
Science Fiction
Tennis

2019-01 – 2019-11

Post-Doctoral Research Associate

School of Computing and Information • University of Pittsburgh

- Building a novel probabilistic relational agent-based modeling-and-simulation framework
- Building Web front ends to an automated modeling and simulation system
- Running disease dynamics and other simulations
- Designing a modeling stack to lower barriers of entry for modelers
- Leading software development and research efforts
- Liaising with clients
- Mentoring; Interviewing candidates
- **Datasets:** Epidemiology, public health, synthetic population

2014-05 – 2018-12

Post-Doctoral Research Associate

Learning and Research Development Center • University of Pittsburgh

- Building a new paradigm distributed data-collection software based on eye movements and end-to-end machine learning pipeline
- Building psychometric and data collection instruments
- Running computational psycholinguistics experiments
- Designing human-subject experiments to test hypotheses
- Handling large datasets
- Performing statistical inference and data analysis
- Authoring research articles
- **Datasets:** Biometric, behavioral, psychometric, computational

2003-09 – 2014-04

Researcher • Software Engineer • Research/Teaching Assistant • PhD Candidate

School of Information Sciences • University of Pittsburgh

- Building ed-tech Web apps that adapt to learner's progress
- Building and deploying Bayesian user models and learner models
- Using eye tracking to study eye movements of readers
- Using eye tracking to evaluate ed-tech learner-adaptive tools
- Designing and running computational experiments with graphical probabilistic models
- Designing and running human-subject experiments to test hypotheses
- Performing statistical inference and data analysis
- Building psychometric and data-collection instruments
- Designing human-robot user interfaces
- Authoring research articles
- **Datasets:** Biometric, behavioral, psychometric, educational, computational

2001-10 – 2003-07

Full-Stack Web Developer • Web Server Administrator

Wroclaw University of Science and Technology • Poland

2002-06 – 2002-08

Data Acquisition Engineer

CIT Engineering • Belgium (Leonardo da Vinci program)

- Developing solutions to data acquisition problems using LabVIEW
- Designing data structures and Web user interface for a distributed data acquisition software
- **Datasets:** Industrial, sensor

Research Experience

tomekloboda.net/research-exp